



## Halton Design and Pareto-Based Sensitivity Analysis

**Purpose:** The purpose of this practical assignment is to reproduce the sensitivity analysis methodology presented by Mintis (2026) using the Isalos Analytics Platform and the SimpleBox4nano web application. A Halton Design is used to generate input scenarios for 13 physicochemical parameters. The simulation responses are then analysed in Isalos using Pareto-based analysis to identify the most influential parameters and their second-order effects.

**Isalos version used:** 2.2.0

### Scientific article:

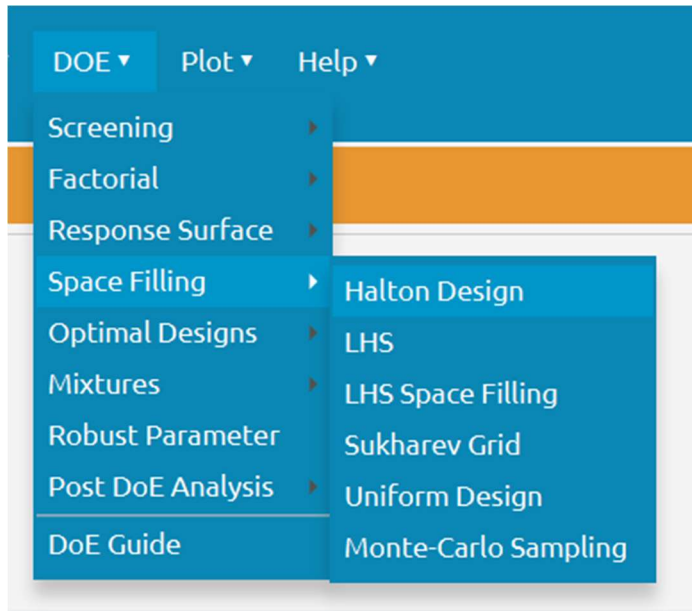
Mintis, D. G., et al. (2026). *SimpleBox4nano: environmental fate modelling of nanomaterials via the Enalos Cloud Platform*. RSC Sustainability, 4, 3147–3164. DOI: 10.1039/D6SU00092D.

### Step 1: Halton Design Generation

Rename the first spreadsheet from “Action” to “Halton Design”. Then, go to DOE → Space Filling → Halton Design. Define the 13 physicochemical parameters using the lower and upper bounds reported in Table 2 of the paper. Set the number of samples to 10,000 and click Execute to generate the Halton design.

The generated design contains 10,000 combinations of the 13 input parameters, providing systematic coverage of the multidimensional parameter space.

| Col1        | Col2 (I) | Col3 (I) | Col4 (I) | Col5 (D) | Col6 (D) | Col7 (I) | Col8 (D) | Col9 (D) | Col10 (D) | Col11 (D) | Col12 (D)          | Col13 (D)            | Col14 (D)       | Col15 |
|-------------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|-----------|--------------------|----------------------|-----------------|-------|
| User Row ID | Radius   | Density  | pKa      | Pvap     | Sol      | Tm       | Kaw      | Kow      | Ksw       | Hamaker   | TransformationRate | AttachmentEfficiency | DissolutionRate |       |
| 1           |          | 900      | -2       | 0.1      | 0.1      | 15       | 1.0E-20  | 1.0E-20  | 0.5       | 1.0E-21   | 1.0E-20            | 1.0E-10              | 1.0E-6          |       |
| 100         |          | 30000    | 14       | 1000000  | 1000000  | 4000     | 100      | 10000000 | 100000    | 1.0E-19   | 0.001              | 1                    | 0.001           |       |



DoE Halton Design

?

×

Number of Center Points per Block

Number of Replicates

Number of Blocks

Number of Samples

☐ Random Standard order

Excluded Columns

Included Columns

Col2 -- Radius  
Col3 -- Density  
Col4 -- pKa  
Col5 -- Pvap  
Col6 -- Sol  
Col7 -- Tm  
Col8 -- Kaw  
Col9 -- Kow  
Col10 -- Ksw

>>

>

<

<<

Execute

Cancel

Results (right spreadsheet) :

| Output      |              | Col1           | Col2 (I)     | Col3 (S)         | Col4 (S)     | Col5 (S) | Col6 (D)    | Col7 (D)   | Col8 (D)     | Col9 (D)     | Col10 (D)    | Col11 (D)  | Col12 (D)    | Col13 (D)    | Col14 (D) | Col15 (D)      | Col16 (D)            | Col17 (D)     | Col18 (D) |
|-------------|--------------|----------------|--------------|------------------|--------------|----------|-------------|------------|--------------|--------------|--------------|------------|--------------|--------------|-----------|----------------|----------------------|---------------|-----------|
| User Header | User Row (D) | Standard Order | Block Number | Replicate Number | Point Type   | Radius   | Density     | pKa        | Prp          | Sol          | Tm           | Kow        | Kow          | Kow          | Hamaker   | TransformaRate | AttachmentEfficiency | DispositioRak |           |
|             | 1            | 1              | Block 1      | Replicate 1      | Design Point | 50.5     | 10600.0     | 1.2000000  | 142857.22857 | 90909.181818 | 321.5384615  | 5.8823529  | 526315.78947 | 4348.3043478 | 0E-7      | 0.0000323      | 0.0270270            | 0.000254      |           |
|             | 2            | 2              | Block 1      | Replicate 1      | Design Point | 25.75    | 20300.0     | 4.4        | 28571.435714 | 181818.26363 | 628.0769231  | 11.7647059 | 1052631.5789 | 8696.1066957 | 0E-7      | 0.00002645     | 0.0540541            | 0.000497      |           |
|             | 3            | 3              | Block 1      | Replicate 1      | Design Point | 75.25    | 4133.333333 | 7.6000000  | 428571.48571 | 272727.34545 | 93461.53846  | 17.6470588 | 1578947.3684 | 13043.913043 | 0E-7      | 0.0000968      | 0.0810811            | 0.000741      |           |
|             | 4            | 4              | Block 1      | Replicate 1      | Design Point | 13.375   | 13833.33333 | 10.8       | 571428.61428 | 363636.42727 | 1241.1538462 | 23.5294118 | 2105263.1578 | 17991.717391 | 0E-7      | 0.0001290      | 0.1081081            | 0.000985      |           |
|             | 5            | 5              | Block 1      | Replicate 1      | Design Point | 62.875   | 23533.33333 | 3          | 714285.74285 | 454545.50909 | 1547.6923077 | 28.4117647 | 2431578.9473 | 21739.521739 | 0E-7      | 0.0001613      | 0.1351351            | 0.001228      |           |
|             | 6            | 6              | Block 1      | Replicate 1      | Design Point | 38.125   | 7366.666667 | 1.8400000  | 857142.87142 | 545454.59090 | 1854.2307692 | 35.2941176 | 3157894.7368 | 26087.326087 | 0E-7      | 0.0001935      | 0.1621622            | 0.001472      |           |
|             | 7            | 7              | Block 1      | Replicate 1      | Design Point | 87.625   | 17066.66666 | 8.04       | 20408.261224 | 636363.67272 | 2160.7692308 | 41.1764706 | 3684210.5263 | 30435.130434 | 0E-7      | 0.0002258      | 0.1891892            | 0.001716      |           |
|             | 8            | 8              | Block 1      | Replicate 1      | Design Point | 7.1875   | 24766.66666 | 7          | 143265.38979 | 272727.75454 | 2467.3076923 | 47.0588235 | 4210526.3157 | 34782.934783 | 0E-7      | 0.0002581      | 0.2162162            | 0.001959      |           |
|             | 9            | 9              | Block 1      | Replicate 1      | Design Point | 56.6875  | 1977.777778 | 11.4400000 | 306122.51836 | 818181.83636 | 2773.8461538 | 52.9411765 | 4736842.1052 | 39130.739130 | 0E-7      | 0.0002903      | 0.2432432            | 0.002203      |           |
|             | 10           | 10             | Block 1      | Replicate 1      | Design Point | 31.9375  | 11677.77777 | 2.72       | 448979.64893 | 909090.91818 | 3080.3846154 | 58.8235294 | 5283157.8947 | 43478.543478 | 0E-7      | 0.0003226      | 0.2702703            | 0.002447      |           |
|             | 11           | 11             | Block 1      | Replicate 1      | Design Point | 81.4375  | 21377.77777 | 2.4800000  | 591836.77351 | 8264.5619833 | 3386.9230769 | 64.7058824 | 5789473.6842 | 47826.347836 | 0E-7      | 0.0003548      | 0.2972973            | 0.002690      |           |
|             | 12           | 12             | Block 1      | Replicate 1      | Design Point | 19.5625  | 5211.111111 | 5.6800000  | 734693.90408 | 99173.643801 | 3693.4615385 | 70.5882353 | 6315789.4736 | 52174.152173 | 0E-7      | 0.0003871      | 0.3243243            | 0.002934      |           |
|             | 13           | 13             | Block 1      | Replicate 1      | Design Point | 69.0625  | 14911.11111 | 8.88       | 877551.03265 | 190082.72561 | 38.5798017   | 76.4705882 | 6842105.2631 | 56521.956521 | 0E-7      | 0.0004194      | 0.3513514            | 0.003178      |           |
|             | 14           | 14             | Block 1      | Replicate 1      | Design Point | 44.3125  | 24611.11111 | 12.08      | 40816.423449 | 309991.80743 | 345.1183432  | 82.3529412 | 738421.0526  | 60869.760869 | 0E-7      | 0.0004516      | 0.3783784            | 0.003421      |           |
|             | 15           | 15             | Block 1      | Replicate 1      | Design Point | 93.8125  | 8444.444444 | -0.0800000 | 183673.55102 | 371900.88925 | 651.6568047  | 88.2352941 | 7894736.8421 | 65217.565217 | 0E-7      | 0.0004839      | 0.4054054            | 0.003665      |           |

....up to 10 000

## Step 2: Execution of the Halton Scenarios in SimpleBox4nano

After generating the 10,000 input scenarios in Isalos, export the Halton Design dataset and save it in CSV format. Open the SimpleBox4nano web application through the Enalos Cloud Platform

<https://www.enaloscloud.novamechanics.com/insight/simplebox4nano/index.zul> and navigate to the Sensitivity Exploration functionality.

In the Sensitivity Exploration window, select the option to upload a CSV file and import the file containing the 10,000 Halton scenarios. The uploaded file contains the combinations of the 13 physicochemical parameters generated in Step 1. The emission and landscape settings are kept fixed while the physicochemical properties are varied across the different scenarios, following the methodology described in the paper.

After successfully importing the scenarios, click Execute to run the simulations. SimpleBox4nano performs a steady-state simulation for each of the 10,000 scenarios and calculates the corresponding environmental fate outputs. The generated outputs include the masses of the solid (S), hetero-aggregated (A), attached (P), and dissolved/gaseous (G/D) species.

Once all simulations are completed, download the results in Excel format. These results are used as the response data for the subsequent Pareto-based sensitivity analysis in Isalos. The paper also specifies that the total mass of solids (S + A + P) is used as the response for the sensitivity analysis.

Sensitivity Exploration

[Add Row](#)
[Remove Selected](#)
[Upload CSV](#)
[Download CSV Template](#)
[Clear](#)
[Execute](#)
[Download Results \(Excel\)](#)

Input scenarios (Diameter → DissRate). Add rows manually or upload CSV.

| Row                              | RadS (n) | Density | pKa  | Pvap (P) | Sol (mg) | Tm (°C) | Kaw  | Kow | Ksw | Hamake | TransRz | AttachEi | DissRate | D (kg)  | S (kg)  | A (kg)  | P (kg)  |
|----------------------------------|----------|---------|------|----------|----------|---------|------|-----|-----|--------|---------|----------|----------|---------|---------|---------|---------|
| <input type="radio"/>            | 50.1     | 106     | 1.2  | 142      | 909      | 321     | 5.81 | 526 | 434 | 4.4    | 0.01    | 0.01     | 0.01     | 1.78774 | 9849.26 | 136351  | 1638.50 |
| <input checked="" type="radio"/> | 25.1     | 203     | 4.4  | 285      | 181      | 628     | 11.1 | 105 | 869 | 7.8    | 0.01    | 0.01     | 0.01     | 2.34637 | 5952.38 | 55767.2 | 887.415 |
| <input type="radio"/>            | 75.1     | 413     | 7.6  | 428      | 272      | 934     | 17.1 | 157 | 130 | 1.1    | 0.01    | 0.01     | 0.01     | 2.75942 | 3852.32 | 89969.5 | 524.075 |
| <input type="radio"/>            | 13.1     | 138     | 10.1 | 571      | 363      | 124     | 23.1 | 210 | 173 | 1.4    | 0.01    | 0.11     | 0.01     | 2.98806 | 3037.35 | 30081.4 | 447.439 |
| <input type="radio"/>            | 62.1     | 235     | -1.1 | 714      | 454      | 154     | 29.1 | 263 | 217 | 1.8    | 0.01    | 0.11     | 0.01     | 3.21777 | 2523.87 | 34672.2 | 353.437 |
| <input type="radio"/>            | 38.1     | 736     | 1.8  | 857      | 545      | 185     | 35.1 | 315 | 260 | 2.1    | 0.01    | 0.11     | 0.01     | 3.32042 | 2178.35 | 28038.9 | 297.225 |
| <input type="radio"/>            | 87.1     | 170     | 5.0  | 204      | 636      | 216     | 41.1 | 368 | 304 | 2.4    | 0.01    | 0.11     | 0.01     | 3.48994 | 1807.58 | 30711.3 | 245.740 |

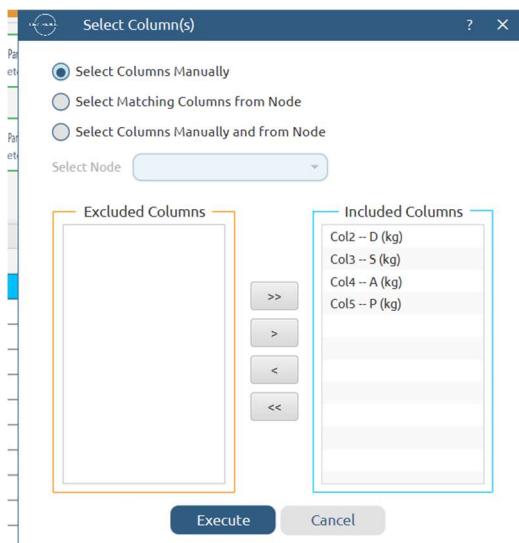
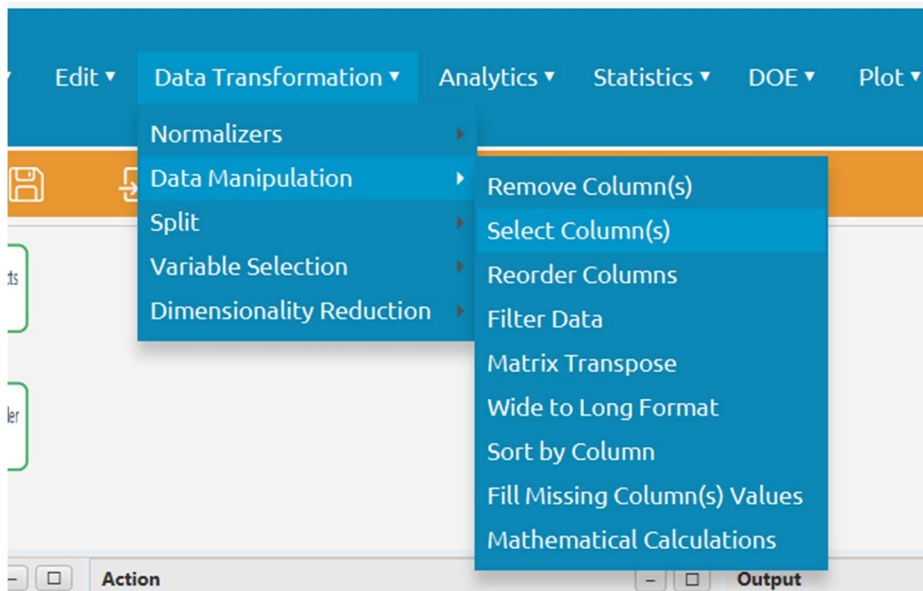
| A                   | B        | C       | D      | E       | F       | G       | H       | I          | J       | K        | L           | M         | N         | O                | P       | Q       | R      |
|---------------------|----------|---------|--------|---------|---------|---------|---------|------------|---------|----------|-------------|-----------|-----------|------------------|---------|---------|--------|
|                     | Diameter | Density | pKa    | Pvap    | Sol     | Tm      | Kaw     | Kow        | Ksw     | Hamaker  | TransRateCc | AttachEff | DissRate  | D (kg)           | S (kg)  | A (kg)  | P (kg) |
| 1                   | 50.5     | 10600   | 1.2    | 142857  | 90909.2 | 321.538 | 5.88235 | 526315.79  | 4348.3  | 4.41E-21 | 0.0000323   | 0.02703   | 0.0000254 | 1.78774e+9849.26 | 136351  | 1638.50 |        |
| 2                   | 25.75    | 20300   | 4.4    | 285714  | 181818  | 628.077 | 11.7647 | 1052631.58 | 8696.11 | 7.83E-21 | 0.0000645   | 0.05405   | 0.0000497 | 2.34637e+5952.38 | 55767.2 | 887.415 |        |
| 3                   | 75.25    | 4133.33 | 7.6    | 428571  | 272727  | 934.615 | 17.6471 | 1578947.37 | 13043.9 | 1.12E-20 | 0.0000968   | 0.08108   | 0.0000741 | 2.75942e+3852.32 | 89969.5 | 524.075 |        |
| 4                   | 13.375   | 13833.3 | 10.8   | 571429  | 363636  | 124.15  | 23.5294 | 2105263.16 | 17391.7 | 1.47E-20 | 0.000129    | 0.10811   | 0.0000985 | 2.98806e+3037.35 | 30081.4 | 447.439 |        |
| 5                   | 62.875   | 23533.3 | -1.36  | 714286  | 454546  | 1547.69 | 29.4118 | 2631578.95 | 21739.5 | 1.81E-20 | 0.000161    | 0.13514   | 0.000123  | 3.21777e+2523.87 | 34672.2 | 353.437 |        |
| 6                   | 38.125   | 7366.67 | 1.84   | 857143  | 545455  | 1854.23 | 35.2941 | 3157894.74 | 26087.3 | 2.15E-20 | 0.000194    | 0.16216   | 0.000147  | 3.32042e+2178.35 | 28038.9 | 297.225 |        |
| 7                   | 87.625   | 17066.7 | 5.04   | 20408.3 | 636364  | 2160.77 | 41.1765 | 3684210.53 | 30435.1 | 2.49E-20 | 0.000226    | 0.18919   | 0.000172  | 3.48994e+1807.58 | 30711.3 | 245.740 |        |
| 8                   | 7.1875   | 26766.7 | 8.24   | 163265  | 727273  | 2467.31 | 47.0588 | 4210526.32 | 34782.9 | 2.83E-20 | 0.000258    | 0.21622   | 0.000196  | 3.53315e+1473.17 | 20757.5 | 216.958 |        |
| 9                   | 56.6875  | 1977.78 | 11.44  | 306123  | 818182  | 2773.85 | 52.9412 | 4736842.11 | 39130.7 | 3.17E-20 | 0.00029     | 0.24324   | 0.000222  | 3.61212e+1479.96 | 28671.8 | 189.754 |        |
| 10                  | 31.9375  | 11677.8 | -0.72  | 448980  | 909091  | 3080.38 | 58.8235 | 5263157.9  | 43478.5 | 3.51E-20 | 0.000323    | 0.27027   | 0.000245  | 3.67479e+1337.34 | 14996.7 | 183.708 |        |
| 11                  | 81.4375  | 21377.8 | 2.48   | 591837  | 8264.56 | 3386.92 | 64.7059 | 5789473.68 | 47826.3 | 3.86E-20 | 0.000355    | 0.2973    | 0.000269  | 3.78331e+1202.20 | 17957.1 | 162.981 |        |
| 12                  | 19.5625  | 5211.11 | 5.68   | 734694  | 99173.6 | 3693.46 | 70.5882 | 6315789.47 | 52174.2 | 4.2E-20  | 0.000387    | 0.32432   | 0.000293  | 3.80657e+1096.48 | 11136.5 | 155.665 |        |
| 13                  | 69.0625  | 14911.1 | 8.88   | 877551  | 190083  | 38.5799 | 76.4706 | 6842105.26 | 56522   | 4.54E-20 | 0.000419    | 0.35135   | 0.000318  | 3.80409e+1037.52 | 18547.4 | 134.566 |        |
| 14                  | 44.3125  | 24611.1 | 12.08  | 40816.4 | 280992  | 345.118 | 82.3529 | 7368421.05 | 60869.8 | 4.88E-20 | 0.000452    | 0.37838   | 0.000342  | 3.81830e+970.434 | 12255.5 | 131.863 |        |
| 15                  | 93.8125  | 8444.44 | -0.08  | 183674  | 371901  | 651.657 | 88.2353 | 7894736.84 | 65217.6 | 5.22E-20 | 0.000484    | 0.40541   | 0.000366  | 3.86431e+893.733 | 22639.3 | 109.118 |        |
| 16                  | 4.99375  | 18144.4 | 3.12   | 326531  | 462610  | 958.195 | 94.1176 | 8421052.63 | 69565.4 | 5.56E-20 | 0.000516    | 0.43243   | 0.000391  | 3.88418e+865.205 | 22926.8 | 103.235 |        |
| 17                  | 53.5938  | 27844.4 | 6.32   | 469388  | 553719  | 1264.73 | 0.34602 | 8947368.42 | 73913.2 | 5.9E-20  | 0.000548    | 0.45946   | 0.000415  | 7.09589e+804.831 | 10283.4 | 109.239 |        |
| 18                  | 28.8438  | 3055.56 | 9.52   | 612245  | 644628  | 1571.27 | 6.22837 | 9473684.21 | 78261   | 6.24E-20 | 0.000581    | 0.48649   | 0.00044   | 4.10386e+751.577 | 9510.45 | 104.058 |        |
| 19                  | 78.3438  | 12765.6 | 12.72  | 755102  | 735637  | 1877.81 | 12.1107 | 27700.831  | 82808.8 | 6.59E-20 | 0.000613    | 0.51351   | 0.000484  | 3.99544e+718.674 | 14445.0 | 91.1257 |        |
| 20                  | 16.4688  | 22455.6 | 0.56   | 897959  | 826446  | 2184.35 | 17.9931 | 554016.621 | 89956.6 | 6.93E-20 | 0.000645    | 0.54054   | 0.000489  | 3.97678e+659.110 | 5919.31 | 94.5836 |        |
| 21                  | 65.9688  | 6288.89 | 3.76   | 11224.6 | 917355  | 2490.89 | 23.8754 | 1080332.41 | 91304.4 | 7.27E-20 | 0.000677    | 0.56757   | 0.000513  | 3.99408e+655.413 | 14117.5 | 82.1468 |        |
| 22                  | 41.2188  | 15988.9 | 6.96   | 204082  | 16529   | 2797.43 | 29.7578 | 1606648.2  | 95652.2 | 7.61E-20 | 0.00071     | 0.59459   | 0.000537  | 3.99738e+624.321 | 8214.96 | 84.5059 |        |
| 23                  | 90.7188  | 25688.9 | 10.16  | 346939  | 107438  | 3103.96 | 35.6401 | 2132963.99 | 189.535 | 7.95E-20 | 0.000742    | 0.62162   | 0.000561  | 3.21167e+594.036 | 7564.85 | 82.8123 |        |
| 24                  | 10.2813  | 9522.22 | 13.36  | 489796  | 198347  | 3410.5  | 41.5225 | 2659279.78 | 4537.34 | 8.29E-20 | 0.000774    | 0.64865   | 0.000586  | 2.08352e+528.519 | 6880.55 | 76.9907 |        |
| 25                  | 58.7813  | 19222.2 | -1.872 | 632653  | 289256  | 3717.04 | 47.4048 | 3185595.57 | 8885.14 | 8.63E-20 | 0.000806    | 0.67568   | 0.00061   | 2.75271e+553.093 | 8551.71 | 73.1480 |        |
| 26                  | 26.25    | 10000   | 1.5    | 150000  | 100000  | 300.000 | 6.66667 | 266666.667 | 20000.0 | 4.91E-21 | 0.0000400   | 0.03000   | 0.0000300 | 1.99999e+9999.99 | 133333  | 1666.67 |        |
| Significant Results |          |         |        |         |         |         |         |            |         |          |             |           |           |                  |         |         |        |

## Step 3: Definition of Response Variables

Create a new spreadsheet and rename it “Responses”. Enter or import the simulation outputs obtained from the SimpleBox4nano Sensitivity Exploration for the 10,000 Halton scenarios. The four response variables are D (dissolved/gaseous), S (solid), A (hetero-aggregated), and P (attached).

After entering the data, go to Data Transformation → Data Manipulation → Select Column(s) and select the four response columns D, S, A, and P. Click Execute to complete the response dataset.

| Input       |             |            |          |          |          |      |
|-------------|-------------|------------|----------|----------|----------|------|
|             | Col1        | Col2 (D)   | Col3 (D) | Col4 (D) | Col5 (D) | Col6 |
| User Header | User Row ID | D (kg)     | S (kg)   | A (kg)   | P (kg)   |      |
| 1           |             | 1.78774E10 | 9849.26  | 136351   | 1638.5   |      |
| 2           |             | 2.34637E10 | 5952.38  | 55767.2  | 887.415  |      |
| 3           |             | 2.75942E10 | 3852.32  | 89969.5  | 524.075  |      |
| 4           |             | 2.98806E10 | 3037.35  | 30081.4  | 447.439  |      |
| 5           |             | 3.21777E10 | 2523.87  | 34672.2  | 353.437  |      |
| 6           |             | 3.32042E10 | 2178.35  | 28038.9  | 297.225  |      |
| 7           |             | 3.48994E10 | 1807.58  | 30711.3  | 245.74   |      |
| 8           |             | 3.53315E10 | 1473.17  | 20757.5  | 216.958  |      |
| 9           |             | 3.61212E10 | 1479.96  | 28671.8  | 189.754  |      |
| 10          |             | 3.67479E10 | 1337.34  | 14996.7  | 183.708  |      |
| 11          |             | 3.78331E10 | 1202.2   | 17957.1  | 162.981  |      |
| 12          |             | 3.80657E10 | 1096.48  | 11136.5  | 155.665  |      |
| 13          |             | 3.80409E10 | 1037.52  | 18547.4  | 134.566  |      |
| 14          |             | 3.8183E10  | 970.434  | 12255.5  | 131.863  |      |
| 15          |             | 3.86431E10 | 893.733  | 22639.3  | 109.118  |      |



| Output      |             |             |          |          |          |
|-------------|-------------|-------------|----------|----------|----------|
|             | Col1        | Col2 (D)    | Col3 (D) | Col4 (D) | Col5 (D) |
| User Header | User Row ID | D (kg)      | S (kg)   | A (kg)   | P (kg)   |
| 1           |             | 1.78774E+10 | 9849.26  | 136351   | 1638.5   |
| 2           |             | 2.34637E+10 | 5952.38  | 55767.2  | 887.415  |
| 3           |             | 2.75942E+10 | 3852.32  | 89969.5  | 524.075  |
| 4           |             | 2.98806E+10 | 3037.35  | 30081.4  | 447.439  |
| 5           |             | 3.21777E+10 | 2523.87  | 34672.2  | 353.437  |
| 6           |             | 3.32042E+10 | 2178.35  | 28038.9  | 297.225  |
| 7           |             | 3.48994E+10 | 1807.58  | 30711.3  | 245.74   |
| 8           |             | 3.53315E+10 | 1473.17  | 20757.5  | 216.958  |
| 9           |             | 3.61212E+10 | 1479.96  | 28671.8  | 189.754  |
| 10          |             | 3.67479E+10 | 1337.34  | 14996.7  | 183.708  |
| 11          |             | 3.78331E+10 | 1202.2   | 17957.1  | 162.981  |
| 12          |             | 3.80657E+10 | 1096.48  | 11136.5  | 155.665  |
| 13          |             | 3.80409E+10 | 1037.52  | 18547.4  | 134.566  |
| 14          |             | 3.8183E+10  | 970.434  | 12255.5  | 131.863  |
| 15          |             | 3.86431E+10 | 893.733  | 22639.3  | 109.118  |
| 16          |             | 3.88418E+10 | 695.205  | 22926.8  | 103.235  |
| 17          |             | 7.09569E+10 | 804.831  | 10283.4  | 109.239  |
| 18          |             | 4.10366E+10 | 751.577  | 8510.45  | 104.058  |
| 19          |             | 3.99544E+10 | 718.674  | 14445.0  | 91.1257  |
| 20          |             | 3.97678E+10 | 659.11   | 5819.31  | 94.5836  |
| 21          |             | 3.99408E+10 | 655.413  | 14117.5  | 82.1468  |
| 22          |             | 3.99738E+10 | 624.321  | 8214.96  | 84.5059  |

## Step 4: Mathematical Calculation

Create a new spreadsheet and rename it “Addition”. Import the Responses spreadsheet containing the four response variables D, S, A, and P.

Then, go to Data Transformation → Mathematical Calculations → Addition. In the Mathematic Calculations window, click Add under *Calculation Variables* and name the new calculation variable “Total\_Solid”. Select the columns S, A, and P and use the addition operator (+) to define the calculation:

$$\text{Total\_Solid} = S + A + P$$

Click Execute to calculate the total solid mass for all scenarios.

|       |         |         |  |  |
|-------|---------|---------|--|--|
| 37.35 | 30081.4 | 447.439 |  |  |
| 23.87 | 34672.2 | 353.437 |  |  |
| 78.35 | 28038.9 |         |  |  |
| 07.58 | 30711.3 |         |  |  |
| 73.17 | 20757.5 |         |  |  |
| 79.96 | 28671.8 |         |  |  |
| 37.34 | 14996.7 |         |  |  |
| 02.2  | 17957.1 |         |  |  |
| 96.48 | 11136.5 |         |  |  |
| 37.52 | 18547.4 |         |  |  |
| 0.434 | 12255.5 |         |  |  |
| 3.733 | 22639.3 |         |  |  |

- Show Spreadsheet Toolbar
- Import from File
- Import from Database
- Import from Spreadsheet
- Re-Import from Spreadsheet
- Import from Multiple Spreadsheets
- Adjust Spreadsheet Precision
- Export Spreadsheet Data
- Clear Spreadsheet

Choose tab as input

Select input tab

Responses

Execute

Cancel

Edit

Data Transformation

Analytics

Statistics

DOE

Plot

Normalizers

Data Manipulation

Split

Variable Selection

Dimensionality Reduction

Remove Column(s)

Select Column(s)

Reorder Columns

Filter Data

Matrix Transpose

Wide to Long Format

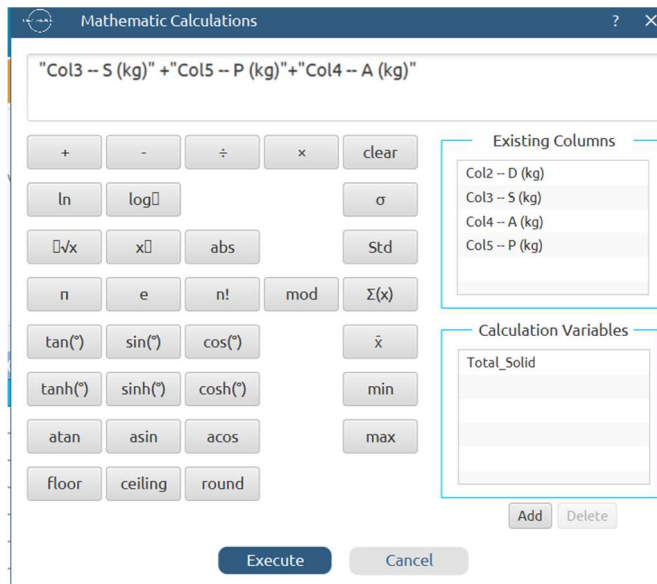
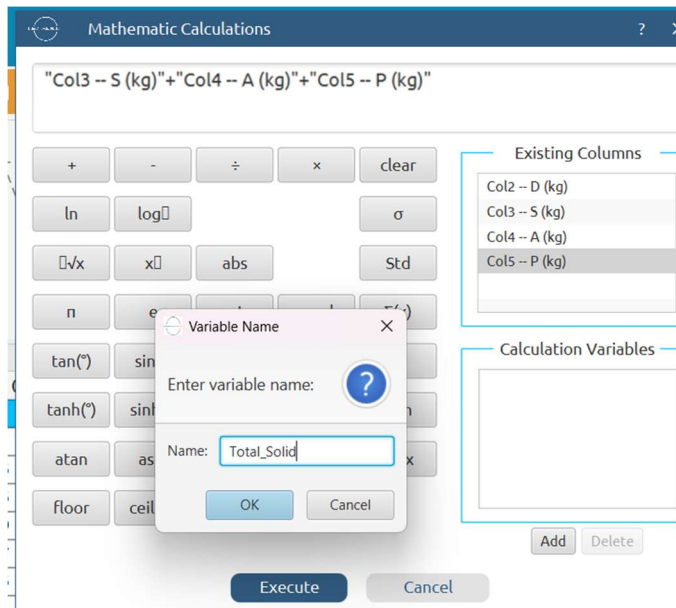
Sort by Column

Fill Missing Column(s) Values

Mathematical Calculations

Action

Output



| Output      |             |             |          |          |          |               |
|-------------|-------------|-------------|----------|----------|----------|---------------|
|             | Col1        | Col2 (D)    | Col3 (D) | Col4 (D) | Col5 (D) | Col6 (D)      |
| User Header | User Row ID | D (kg)      | S (kg)   | A (kg)   | P (kg)   | Total_Solid   |
| 1           |             | 1.78774E+10 | 9849.26  | 136351   | 1638.5   | 147838.76     |
| 2           |             | 2.34637E+10 | 5952.38  | 55767.2  | 887.415  | 62606.9950000 |
| 3           |             | 2.75942E+10 | 3852.32  | 89969.5  | 524.075  | 94345.895     |
| 4           |             | 2.98806E+10 | 3037.35  | 30081.4  | 447.439  | 33566.189     |
| 5           |             | 3.21777E+10 | 2523.87  | 34672.2  | 353.437  | 37549.507     |
| 6           |             | 3.32042E+10 | 2178.35  | 28038.9  | 297.225  | 30514.4750000 |
| 7           |             | 3.48994E+10 | 1807.58  | 30711.3  | 245.74   | 32764.62      |
| 8           |             | 3.53315E+10 | 1473.17  | 20757.5  | 216.958  | 22447.628     |
| 9           |             | 3.61212E+10 | 1479.96  | 28671.8  | 189.754  | 30341.514     |
| 10          |             | 3.67479E+10 | 1337.34  | 14996.7  | 183.708  | 16517.748     |
| 11          |             | 3.78331E+10 | 1202.2   | 17957.1  | 162.981  | 19322.281     |
| 12          |             | 3.80657E+10 | 11096.48 | 11136.5  | 155.665  | 12388.645     |
| 13          |             | 3.80409E+10 | 1037.52  | 18547.4  | 134.566  | 19719.486     |
| 14          |             | 3.8183E+10  | 970.434  | 12255.5  | 131.863  | 13357.797     |
| 15          |             | 3.86431E+10 | 893.733  | 22639.3  | 109.118  | 23642.1510000 |
| 16          |             | 3.88418E+10 | 695.205  | 22926.8  | 103.235  | 23725.2400000 |
| 17          |             | 7.09569E+10 | 804.831  | 10283.4  | 109.239  | 11197.47      |

## Step 5: Total Solid

Create a new spreadsheet and rename it "Total Solid". Import the output obtained from the Addition spreadsheet.

Then, go to Data Transformation → Data Manipulation → Select Column(s). Keep D (kg), S (kg), A (kg), P (kg) and Total\_Solid as the Included Columns. Click Execute to complete the selection.

|   |         |         |         | (kg)"        |   |
|---|---------|---------|---------|--------------|---|
| 0 | 9849.26 | 136351  | 1638.5  | 147838.76    |   |
| 0 | 5952.38 | 55767.2 | 887.415 | 62606.995000 | 0 |
| 0 | 3852.32 | 8996    |         |              |   |
| 0 | 3037.35 | 3008    |         |              |   |
| 0 | 2523.87 | 3467    |         |              |   |
| 0 | 2178.35 | 2808    |         |              |   |
| 0 | 1807.58 | 3078    |         |              |   |
| 0 | 1473.17 | 2078    |         |              |   |
| 0 | 1479.96 | 2868    |         |              |   |
| 0 | 1337.34 | 1498    |         |              |   |
| 0 | 1202.2  | 1798    |         |              |   |
| 0 | 1096.48 | 1118    |         |              |   |

Show Spreadsheet Toolbar

Import from File

Import from Database

Import from Spreadsheet

Re-Import from Spreadsheet

Import from Multiple Spreadsheets

Adjust Spreadsheet Precision

Export Spreadsheet Data

Clear Spreadsheet

Choose tab as input

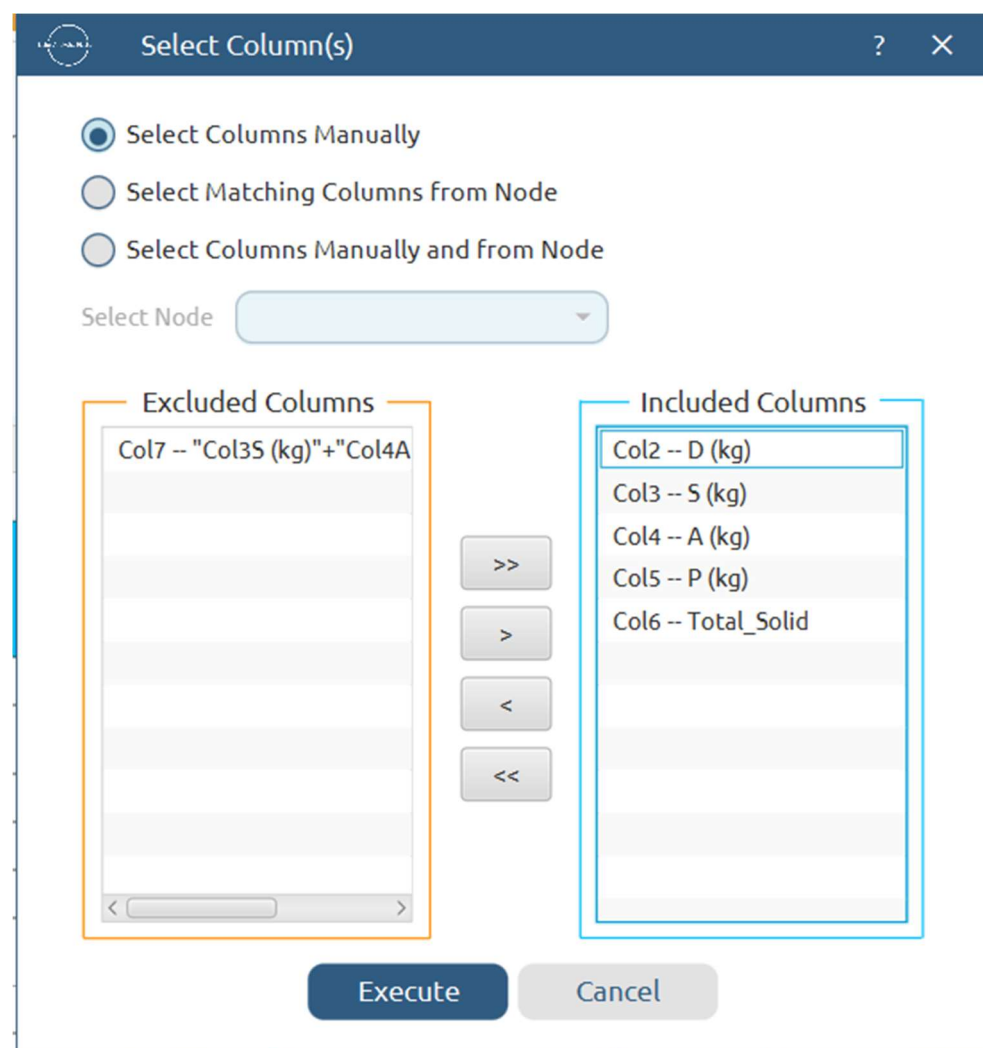
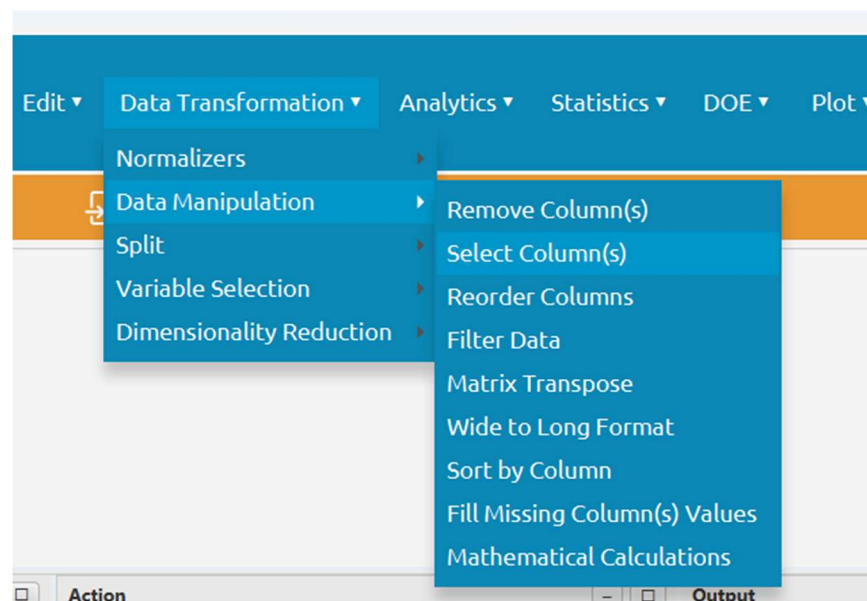
×

Select input tab

Addition

Execute

Cancel



| Output      |             |             |          |          |          |               |     |
|-------------|-------------|-------------|----------|----------|----------|---------------|-----|
|             | Col1        | Col2 (D)    | Col3 (D) | Col4 (D) | Col5 (D) | Col6 (D)      | Col |
| User Header | User Row ID | D (kg)      | S (kg)   | A (kg)   | P (kg)   | Total_Solid   |     |
| 1           |             | 1.78774E+10 | 9849.26  | 136351   | 1638.5   | 147838.76     |     |
| 2           |             | 2.34637E+10 | 5952.38  | 55767.2  | 887.415  | 62606.9950000 |     |
| 3           |             | 2.75942E+10 | 3852.32  | 89969.5  | 524.075  | 94345.895     |     |
| 4           |             | 2.98806E+10 | 3037.35  | 30081.4  | 447.439  | 33566.189     |     |
| 5           |             | 3.21777E+10 | 2523.87  | 34672.2  | 353.437  | 37549.507     |     |
| 6           |             | 3.32042E+10 | 2178.35  | 28038.9  | 297.225  | 30514.4750000 |     |
| 7           |             | 3.48994E+10 | 1807.58  | 30711.3  | 245.74   | 32764.62      |     |
| 8           |             | 3.53315E+10 | 1473.17  | 20757.5  | 216.958  | 22447.628     |     |
| 9           |             | 3.61212E+10 | 1479.96  | 28671.8  | 189.754  | 30341.514     |     |
| 10          |             | 3.67479E+10 | 1337.34  | 14996.7  | 183.708  | 16517.748     |     |
| 11          |             | 3.78331E+10 | 1202.2   | 17957.1  | 162.981  | 19322.281     |     |
| 12          |             | 3.80657E+10 | 1096.48  | 11136.5  | 155.665  | 12388.645     |     |
| 13          |             | 3.80409E+10 | 1037.52  | 18547.4  | 134.566  | 19719.486     |     |
| 14          |             | 3.8183E+10  | 970.434  | 12255.5  | 131.863  | 13357.797     |     |
| 15          |             | 3.86431E+10 | 893.733  | 22639.3  | 109.118  | 23642.1510000 |     |
| 16          |             | 3.88418E+10 | 695.205  | 22926.8  | 103.235  | 23725.2400000 |     |
| 17          |             | 7.09569E+10 | 804.831  | 10283.4  | 109.239  | 11197.47      |     |

## Step 6: Data Preparation

Create a new spreadsheet and rename it “Data”. Import the combined dataset containing the Halton Design spreadsheet and the Total\_Solid spreadsheet.

Then, go to Data Transformation → Data Manipulation → Select Column(s). Include all the input parameters and the calculated Total\_Solid response. Click Execute to create the final dataset for the sensitivity analysis.

|         |                                               |           |             |         |
|---------|-----------------------------------------------|-----------|-------------|---------|
| 1000900 | 0.0010011                                     | 0.0000741 | 2.75942E+10 | 3852.32 |
| 1001290 | Show Spreadsheet Toolbar                      |           |             |         |
| 1001613 | Import from File                              |           |             |         |
| 1001935 | Import from Database                          |           |             |         |
| 1002258 | Import from Spreadsheet                       |           |             |         |
| 1002581 | Import from Multiple Spreadsheets             |           |             |         |
| 1002581 | Re-Import from Multiple Spreadsheets          |           |             |         |
| 1002903 | Reconfigure Import from Multiple Spreadsheets |           |             |         |
| 1003226 | Adjust Spreadsheet Precision                  |           |             |         |
|         | Export Spreadsheet Data                       |           |             |         |

Multiple Spreadsheet Joiner

?

×

Join Configuration Steps

Step 1: Halton Design X Total\_Solid (Horizontal)

Add

Delete

Join Type

Concatenation:  
☒ Horizontal  
☐ Vertical

Join:  
☐ Left Join ☐ Right Join ☐ Inner Join ☐ Full Outer Join

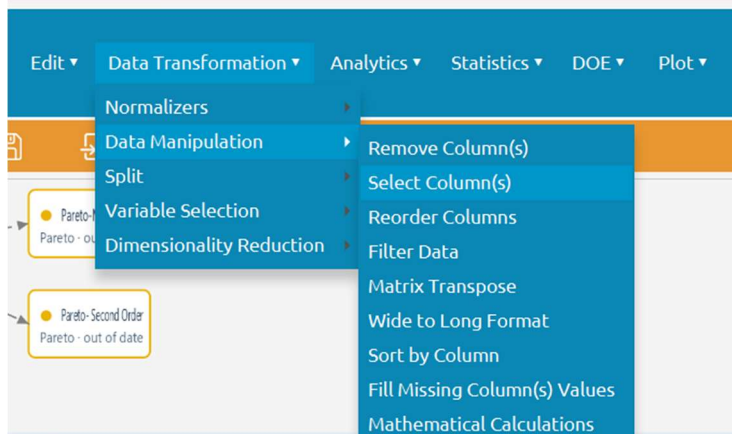
Left Spreadsheet  
Halton Design

Right Spreadsheet  
Total\_Solid

Join Column  
☒ Common header name ☐ Different header names

Execute

Cancel



Select Column(s)

?

×

☒ Select Columns Manually  
☐ Select Matching Columns from Node  
☐ Select Columns Manually and from Node

Select Node

Excluded Columns

Included Columns

>>

>

<

<<

Col2 -- Standard Order

Col3 -- Block Number

Col4 -- Replicate Number

Col5 -- Point Type

Col6 -- Radius

Col7 -- Density

Col8 -- pKa

Col9 -- Pvap

Col10 -- Sol

Col11 -- Tm

Col12 -- Kaw

Execute

Cancel

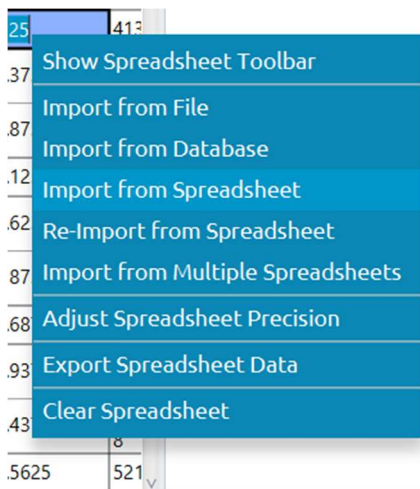
| Output      |             | Col1           | Col2 (I)     | Col3 (S)         | Col4 (S)     | Col5 (S) | Col6 (D)     | Col7 (D)   | Col8 (D)       | Col9 (D)     | Col10 (D)    | Col11 (D)  | Col12 (D)    | Col13 (D)    | Col14 (D)    | Col15 (D)        | Col16 (D)         | Col17 (D)  | Col18 (D) | Col19 (D)   | Col20 (D)   | Col21 (D) | Col22 (D)   | Col23 (D)     | Col24         |         |               |  |
|-------------|-------------|----------------|--------------|------------------|--------------|----------|--------------|------------|----------------|--------------|--------------|------------|--------------|--------------|--------------|------------------|-------------------|------------|-----------|-------------|-------------|-----------|-------------|---------------|---------------|---------|---------------|--|
| User Header | User Row ID | Standard Order | Block Number | Replicate Number | Point Type   | Radius   | Density      | pKa        | pH             | So           | Tm           | Kaw        | Kow          | Kow          | Hammett      | Transformed Data | Adjusted R-Square | Conclusion | O (kg)    | S (kg)      | A (kg)      | P (kg)    | Total Solid |               |               |         |               |  |
| 1           | 1           | 1              | Block 1      | Replicate 1      | Design Point | 50.5     | 10000.0      | 1.2000000  | 142857.22857   | 90909.181818 | 121.5384615  | 5.8823529  | 52515.76947  | 37           | 4348.3043478 | 06-7             | 0.0000323         | 0.0270270  | 0.0000254 | 1.77744E+10 | 9849.26     | 136551    | 1688.5      | 147838.76     |               |         |               |  |
|             | 2           | 2              | Block 1      | Replicate 1      | Design Point | 25.75    | 20300.0      | 4.4        | 285714.35714   | 181818.26363 | 638.0769231  | 11.7647959 | 1052631.5789 | 474          | 8894.1088957 | 06-7             | 0.0000645         | 0.0540541  | 0.0000487 | 2.34637E+10 | 5953.38     | 55787.2   | 887.415     | 62606.9950000 |               |         |               |  |
| 3           | 3           | 3              | Block 1      | Replicate 1      | Design Point | 75.25    | 4133.3333333 | 7.6000000  | 433571.48571   | 212727.34545 | 934.6153846  | 17.6470588 | 1578847.3584 | 13043.913043 | 211          | 1                | 06-7              | 0.0000968  | 0.0810811 | 0.0000741   | 2.75942E+10 | 3852.32   | 89949.5     | 524.075       | 94345.895     |         |               |  |
|             | 4           | 4              | Block 1      | Replicate 1      | Design Point | 13.375   | 15833.333333 | 10.8       | 571428.61428   | 345454.52727 | 1241.1538462 | 23.5294118 | 2105243.1578 | 17391.717391 | 644          | 1                | 06-7              | 0.0001290  | 0.1081081 | 0.0000985   | 2.98806E+10 | 3037.35   | 30081.4     | 447.459       | 33566.189     |         |               |  |
| 5           | 5           | 5              | Block 1      | Replicate 1      | Design Point | 62.875   | 23533.333333 | 3          | 174285.7142857 | 345454.59009 | 77           | 59         | 1474.6923077 | 29.4117647   | 2481578.9473 | 21739.521739     | 684               | 1          | 06-7      | 0.0001613   | 0.1351351   | 0.0001228 | 3.21777E+10 | 2523.87       | 34672.2       | 353.437 | 37549.507     |  |
|             | 6           | 6              | Block 1      | Replicate 1      | Design Point | 38.125   | 7356.6666667 | 1.8400000  | 858714.28571   | 545454.59009 | 854          | 91         | 1854.2307692 | 35.2941176   | 3157894.7968 | 26087.326087     | 421               | 0          | 06-7      | 0.0001935   | 0.1621622   | 0.0001472 | 3.32042E+10 | 2178.35       | 28038.9       | 297.325 | 30514.4700000 |  |
| 7           | 7           | 7              | Block 1      | Replicate 1      | Design Point | 87.625   | 17056.666666 | 5.04       | 250438.261234  | 435363.67372 | 2160.7692308 | 41.1764706 | 3684210.5283 | 30435.130434 | 158          | 8                | 06-7              | 0.0002258  | 0.1891892 | 0.0001716   | 3.48946E+10 | 1807.58   | 30711.3     | 245.74        | 32764.62      |         |               |  |
|             | 8           | 8              | Block 1      | Replicate 1      | Design Point | 7.1875   | 58766.666666 | 8.2400000  | 163265.38979   | 272727.25454 | 59           | 55         | 2467.3076923 | 47.0589235   | 4210526.3157 | 34792.954792     | 895               | 6          | 06-7      | 0.0002581   | 0.2162162   | 0.0001959 | 3.53315E+10 | 1473.17       | 20757.5       | 216.958 | 22647.628     |  |
| 9           | 9           | 9              | Block 1      | Replicate 1      | Design Point | 56.8875  | 1977.7777778 | 11.4400000 | 306122.51836   | 818181.83636 | 2773.8491538 | 52.9411765 | 4736842.1052 | 39130.739130 | 622          | 4                | 06-7              | 0.0002903  | 0.2434342 | 0.0002203   | 3.61212E+10 | 1479.96   | 28671.8     | 189.754       | 30341.514     |         |               |  |
|             | 10          | 10             | Block 1      | Replicate 1      | Design Point | 31.9375  | 11677.777777 | 0.72       | 448979.64693   | 909090.91818 | 18           | 18         | 2080.3846154 | 58.8235294   | 5263157.8947 | 43478.543478     | 368               | 9          | 06-7      | 0.0003226   | 0.2702703   | 0.0002447 | 3.67479E+10 | 1337.34       | 14996.7       | 183.708 | 16517.748     |  |
| 11          | 11          | 11             | Block 1      | Replicate 1      | Design Point | 81.4375  | 21377.777777 | 2.4800000  | 591836.77551   | 108415619635 | 3588.9230769 | 64.7058924 | 5198473.6841 | 47826.347826 | 105          | 1                | 06-7              | 0.0003548  | 0.2972973 | 0.0002690   | 3.78331E+10 | 1202.2    | 17957.1     | 162.961       | 19922.281     |         |               |  |
|             | 12          | 12             | Block 1      | Replicate 1      | Design Point | 19.5625  | 5211.1111111 | 5.6800000  | 714893.95406   | 99173.643851 | 3993.4615385 | 70.5882353 | 6315709.4736 | 52174.152174 | 842          | 9                | 06-7              | 0.0003871  | 0.3243243 | 0.0002934   | 3.80657E+10 | 1096.48   | 11136.5     | 155.865       | 12388.645     |         |               |  |
| 13          | 13          | 13             | Block 1      | Replicate 1      | Design Point | 89.0625  | 14911.111111 | 8.88       | 877551.03265   | 190082.72561 | 38.5798017   | 76.4705882 | 6842105.2631 | 54521.956521 | 579          | 7                | 06-7              | 0.0004194  | 0.3513514 | 0.0003178   | 3.80409E+10 | 1037.52   | 18547.4     | 134.566       | 19719.486     |         |               |  |
|             | 14          | 14             | Block 1      | Replicate 1      | Design Point | 44.3125  | 24611.111111 | 12.08      | 43816.423449   | 280991.80743 | 40           | 40         | 145.1183432  | 82.3529412   | 7388421.0526 | 60869.760869     | 316               | 8          | 06-7      | 0.0004516   | 0.3788784   | 0.0003421 | 3.8183E+10  | 970.434       | 12355.5       | 131.863 | 13357.797     |  |
| 15          | 15          | 15             | Block 1      | Replicate 1      | Design Point | 93.8125  | 8444.4444444 | -0.0800000 | 183673.55102   | 371900.88925 | 54           | 54         | 651.6560047  | 88.2352941   | 7884736.8421 | 65217.565217     | 933               | 4          | 06-7      | 0.0004839   | 0.4054054   | 0.0003685 | 3.88431E+10 | 893.733       | 22839.3       | 105.118 | 23642.151000  |  |
|             | 16          | 16             | Block 1      | Replicate 1      | Design Point | 4.09375  | 18144.444444 | 4.12       | 326530.87949   | 482309.97187 | 958.1932663  | 94.1176471 | 8451052.6315 | 69565.369565 | 759          | 2                | 06-7              | 0.0005161  | 0.4324324 | 0.0003909   | 3.88418E+10 | 695.205   | 22928.8     | 103.235       | 23725.2400000 |         |               |  |
| 17          | 17          | 17             | Block 1      | Replicate 1      | Design Point | 53.59375 | 27844.444444 | 6.3200000  | 489387.80816   | 153719.05289 | 31           | 29         | 1264.7337278 | 0.3460208    | 8847388.4210 | 73913.173913     | 528               | 0          | 06-7      | 0.0005484   | 0.4594995   | 0.0004152 | 3.70969E+10 | 804.831       | 10283.4       | 106.239 | 11197.47      |  |
|             | 18          | 18             | Block 1      | Replicate 1      | Design Point | 28.84375 | 3055.5555556 | 9.5200000  | 612344.93873   | 644628.13471 | 47           | 47         | 1571.3721853 | 6.2283737    | 9473954.2105 | 78280.978280     | 263               | 9          | 06-7      | 0.0005806   | 0.4884885   | 0.0004396 | 4.10366E+10 | 751.577       | 8510.45       | 104.058 | 9386.0850000  |  |

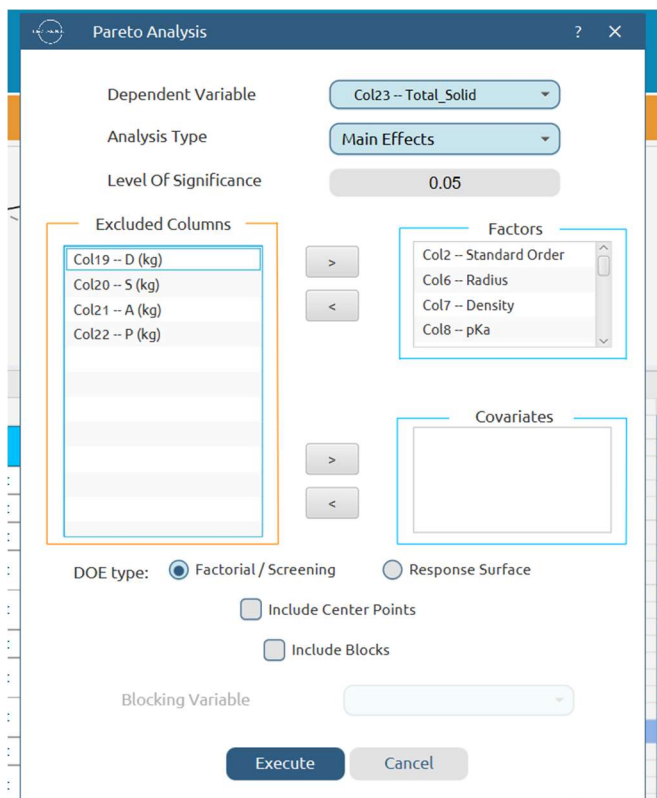
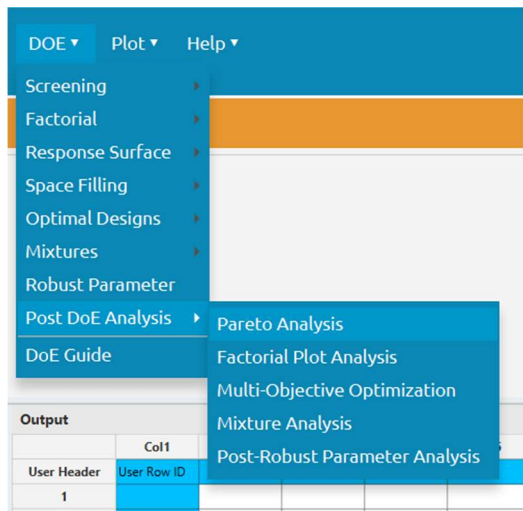
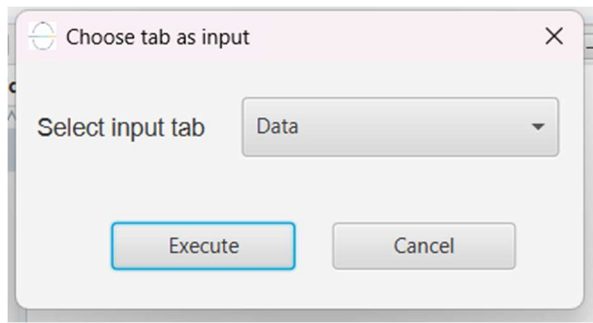
## Step 7: Pareto Analysis – Main Effects

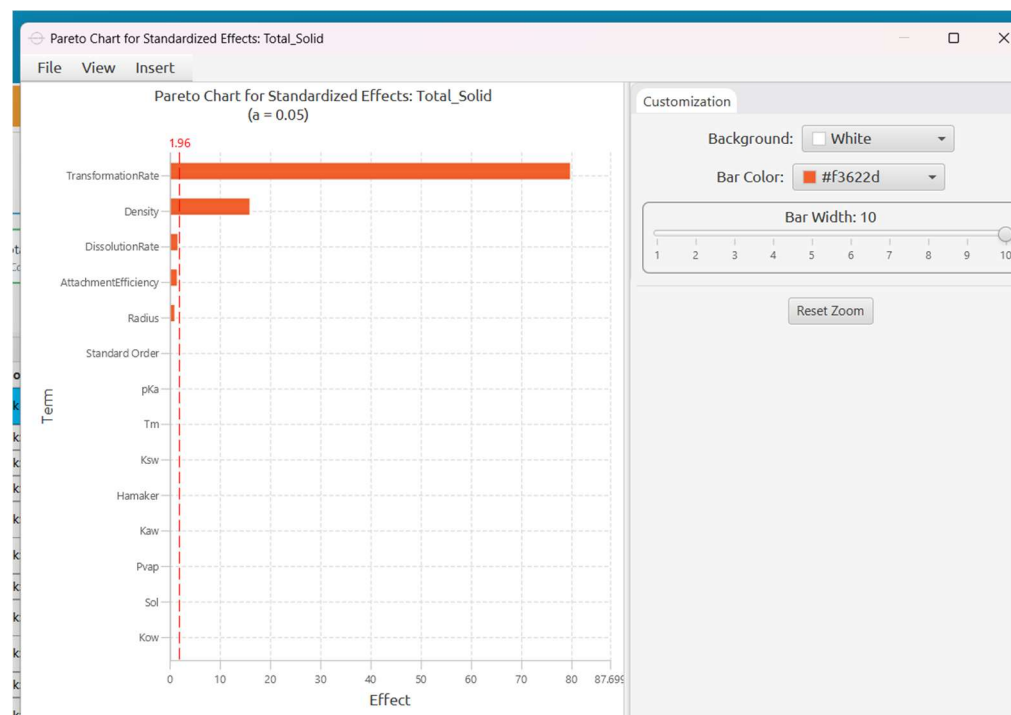
Create a new spreadsheet and rename it “Pareto-Main Effects”. Import the Data spreadsheet.

Go to DOE → Post DOE Analysis → Pareto Analysis. Select Total\_Solid as the Dependent Variable and Main Effects as the Analysis Type. Set the Level of Significance to 0.05.

Select the input parameters as Factors, keep Factorial / Screening as the DOE type, and leave Include Center Points and Include Blocks unchecked. Click Execute to perform the analysis. After the analysis is completed, click Show Chart to display the Pareto chart.







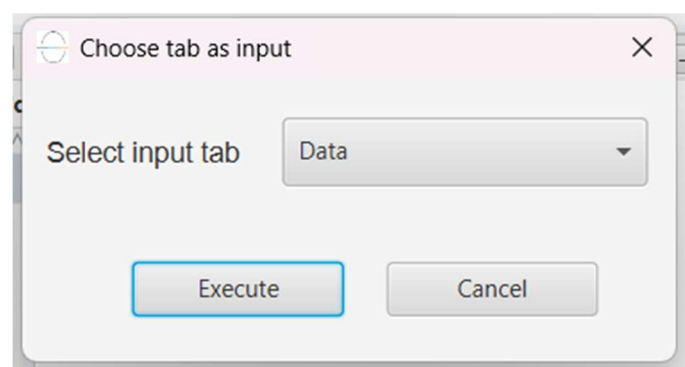
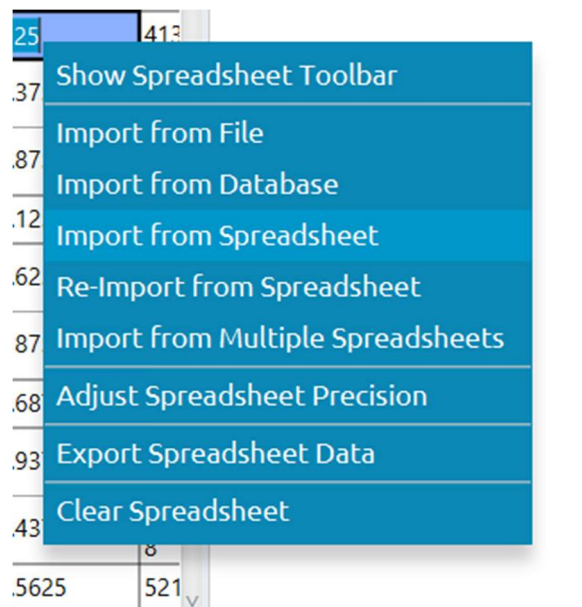
| Output      |             |                      |                      |      |
|-------------|-------------|----------------------|----------------------|------|
|             | Col1        | Col2 (S)             | Col3 (S)             | Col4 |
| User Header | User Row ID | Pareto Analysis of : | Standardized Effects |      |
| 1           |             | Variable             | Effect               |      |
| 2           |             | TransformationRate   | 79.7261167           |      |
| 3           |             | Density              | 15.9033495           |      |
| 4           |             | DissolutionRate      | 1.5651633            |      |
| 5           |             | AttachmentEfficiency | 1.4443781            |      |
| 6           |             | Radius               | 0.9519553            |      |
| 7           |             | Standard Order       | 0.2110044            |      |
| 8           |             | pKa                  | 0.2107462            |      |
| 9           |             | Tm                   | 0.1078256            |      |
| 10          |             | Ksw                  | 0.0609935            |      |
| 11          |             | Hamaker              | 0.0595770            |      |
| 12          |             | Kaw                  | 0.0267575            |      |
| 13          |             | Pvpap                | 0.0248776            |      |
| 14          |             | Sol                  | 0.0206025            |      |
| 15          |             | Kow                  | 0.0149403            |      |
| 16          |             | Significance Value   | 1.9602016            |      |
| 17          |             |                      |                      |      |

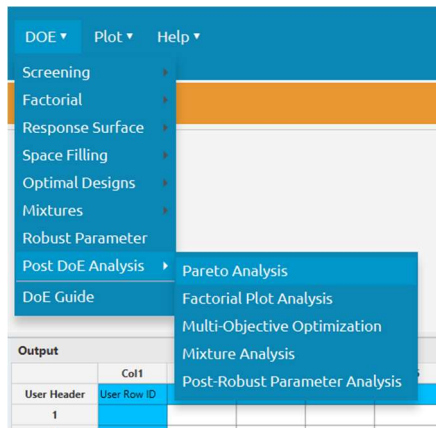
## Step 8: Pareto Analysis – Second Order

Create a new spreadsheet and rename it “Pareto-Second Order”.  
Import the Data spreadsheet.

Go to DOE → Post DOE Analysis → Pareto Analysis. Select Total\_Solid as the Dependent Variable and Main Effects + Two-Factor + Quadratic as the Analysis Type.

Select the input parameters as Factors and choose Response Surface as the DOE type. Leave Include Blocks unchecked and click Execute to perform the second-order analysis. After the analysis is completed, click Show Chart to display the Pareto chart.





Pareto Analysis

Dependent Variable: Col23 -- Total\_Solid

Analysis Type: Main Effects + Two-Factor

Level Of Significance: Double (0,1), Default: 0.05

Excluded Columns: Col19 -- D (kg), Col20 -- S (kg), Col21 -- A (kg), Col22 -- P (kg)

Factors: Col2 -- Standard Order, Col6 -- Radius, Col7 -- Density, Col8 -- pKa

Covariates:

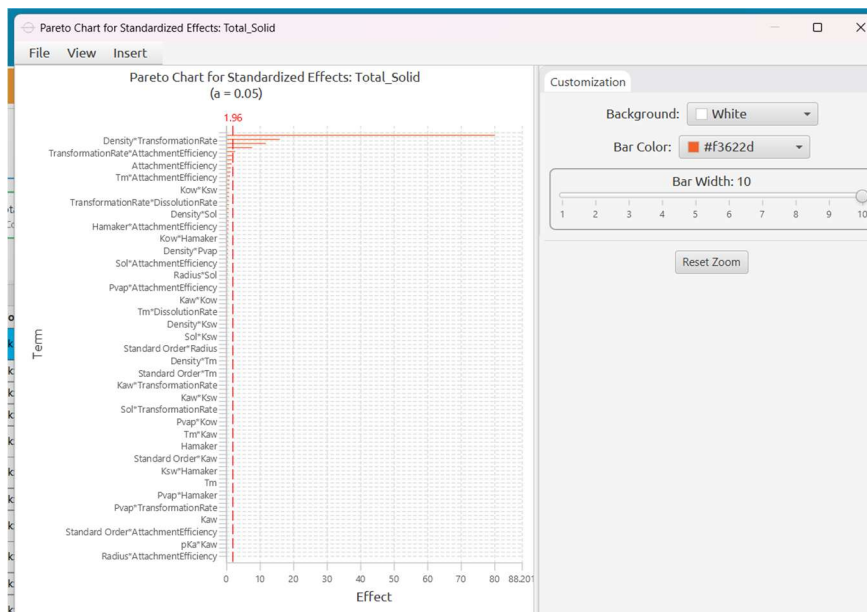
DOE type: ☐ Factorial / Screening ☒ Response Surface

☐ Include Center Points

☐ Include Blocks

Blocking Variable:

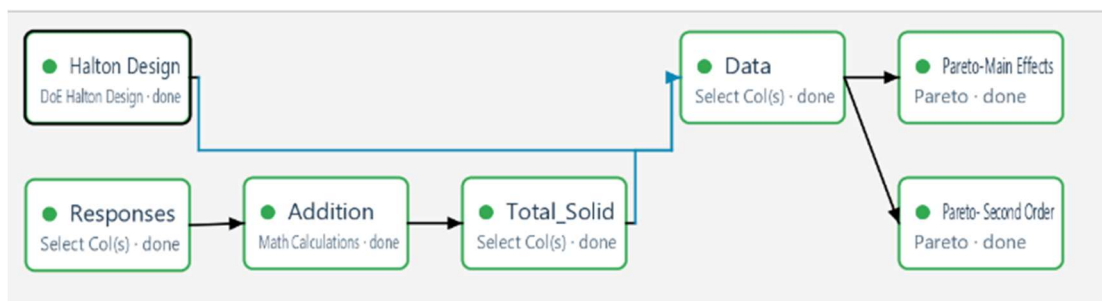
Execute Cancel



| Output      |             |                                         |                      |      |
|-------------|-------------|-----------------------------------------|----------------------|------|
|             | Col1        | Col2 (S)                                | Col3 (S)             | Col4 |
| User Header | User Row ID | Pareto Analysis of :                    | Standardized Effects |      |
| 1           |             | Variable                                | Effect               |      |
| 2           |             | TransformationRate                      | 80.1825391           |      |
| 3           |             | Density                                 | 15.9883443           |      |
| 4           |             | Density*TransformationRate              | 11.8302664           |      |
| 5           |             | Radius*Density                          | 7.6877550            |      |
| 6           |             | Sol*Kaw                                 | 2.6742841            |      |
| 7           |             | TransformationRate*AttachmentEfficiency | 1.7599543            |      |
| 8           |             | DissolutionRate                         | 1.6593107            |      |
| 9           |             | Kaw*AttachmentEfficiency                | 1.5777300            |      |
| 10          |             | AttachmentEfficiency                    | 1.3992716            |      |
| 11          |             | AttachmentEfficiency*DissolutionRate    | 1.2805300            |      |
| 12          |             | Radius*TransformationRate               | 1.1448930            |      |
| 13          |             | Tm*AttachmentEfficiency                 | 0.9669515            |      |

## Final Isalos Workflow

The final workflow is presented below :



## References

Mintis, D. G., Kallishis, S., Kolokathis, P. D., Tsoumanis, A., Melagraki, G., Quik, J. T. K., Meesters, J. A. J., Lynch, I., Greco, D., & Afantitis, A. (2026). *SimpleBox4nano: Environmental fate modelling of nanomaterials via the Enalos Cloud Platform*. RSC Sustainability, 4, 3147–3164.